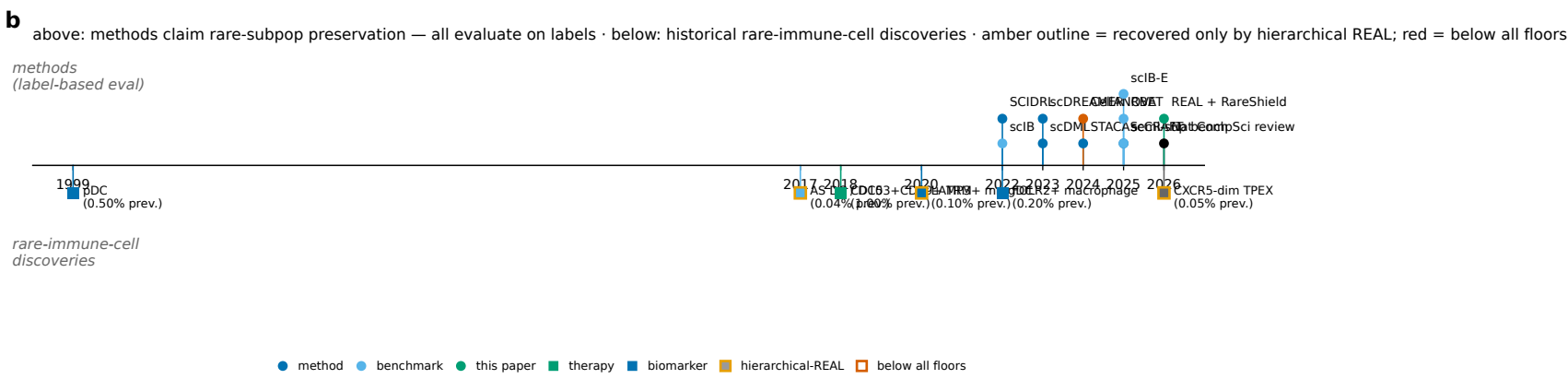
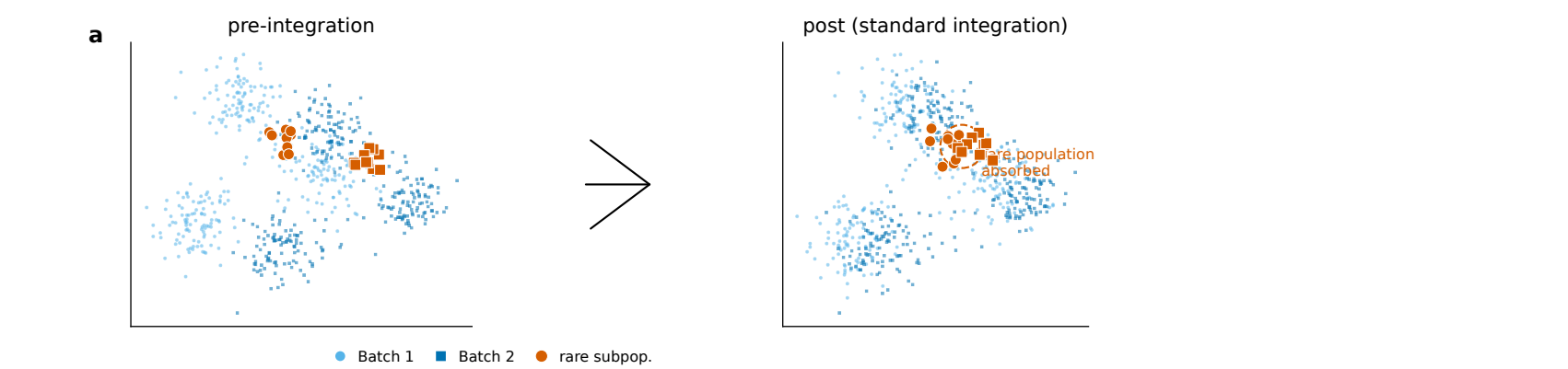
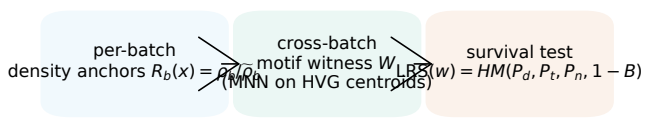


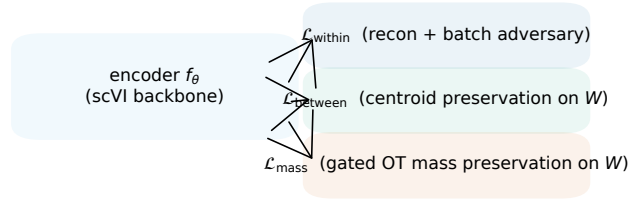


**c** REAL framework: label-free detectability pipeline



label-free, scalable to 4.9M cells via FAISS-GPU + cugraph + witness-complex PH

**d** RareShield: motif-preserving integration

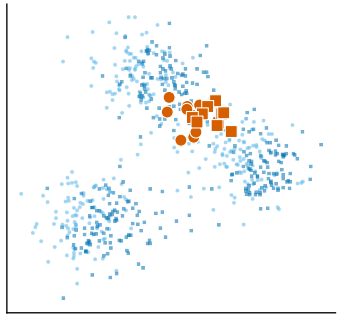


$\mathcal{L}_{mass}$  gated by  $\tau(C_w)^2 \cdot \text{ReLU}(0.4 - \text{LRS}(w))$  (scVI-primary variant)

**e**

standard integrator

LRS = 0.18 (fail)



RareShield

LRS = 0.84 (pass)

